

EWMA-GARCH Hybrid Volatility Forecasting Model

Quant Model Research Report (Sample)

1. Overview

This report describes a daily volatility forecasting model for a single asset. It combines an Exponentially Weighted Moving Average (EWMA) short-term variance estimate with a GARCH(1,1) long-term anchor. The hybrid forecast is a convex combination of the two components and targets daily return variance.

2. Model Formulation

Let r_t be the daily log return. EWMA component: $v_t = \lambda v_{t-1} + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ and v_0 equal to the sample variance. GARCH(1,1) component: $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, with $\omega = 0.00001$, $\alpha = 0.10$, $\beta = 0.85$. The hybrid forecast is $V_t = w v_t + (1 - w) \sigma_t^2$ with mixing weight $w = 0.5$. The model outputs the annualized volatility forecast $\sqrt{V_t \cdot 252}$.

3. Hyper-parameters

EWMA decay $\lambda = 0.94$. GARCH parameters $\omega = 0.00001$, $\alpha = 0.10$, $\beta = 0.85$. Mixing weight $w = 0.5$. Warm-up window of 20 observations is discarded before evaluation.

4. Training and Evaluation

Fit on the first 80% of the return series, forecast the remaining 20% one step ahead. Evaluate with Mean Squared Error between forecast variance and realized squared returns, and with a quasi-log-likelihood QLL = $\text{mean}(-\log(V_t) - r_t^2 / V_t)$. Lower MSE and higher QLL are better.

5. Implementation Notes

Implement in Python using numpy and pandas. Provide a class HybridVolatilityModel with fit(returns) and predict(returns) methods plus a runnable demo script: if no CSV file is provided, generate 500 synthetic daily returns from a normal distribution with mean 0.0005 and std 0.02 (fixed random seed 42), fit the model, and print MSE, QLL and the last annualized volatility forecast.